

A Location Based Hedonic Machine Learning Model to Predict the Impact Factors of House Price Estimates

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1 Introduction

In the real estate industry when it comes to estimating house prices the saying goes that the three most important impact factors are location, location, and location. This statement is true as the location of a house is an important impact factor for its price. At the same time the saying is not true, because House buyers are not willing to pay for the location per se. Instead, the relation between house prices and location is indirect. Perspective homeowners are willing to pay for amenities that come with a location such as a ocean view, a good school nearby, or a low crime rate.

This brings us to hedonic housing price models. The idea is to identify variables and data that are closely related to amenities a house price buyer is willing to pay for. These variables can be location based, such as driving distance from the house to the beach or park, school quality at the nearest highschool, or densities of crimes and restaurants. Non-location-based variables include the size of a house measured in square feet or the number of bedrooms. as they provide the amenities of space.

In a hedonic model that reflect the price impact of all amenities, the price impact of all amenities would sum up to the estimated house price. Realistically, for many amenities like the friendliness of neighbors or the noise level at the location, data are not available. The cumulative influence of all these amenities would be reflected in the constant of a model and the sum measurable amenities would reflect the difference between the constant and the estimated price.

The idea behind a hedonic model is not only to use variables that reflect amenities for estimating the price. A hedonic model, after the price is estimated must also identify the price impact of all its variable, i.e. all its amenities.

To reach the prediction goal of a hedonic model and to quantify the impact factors the analysis proceeds in three steps

1. Geographic Information Systems are used to quantify location related variables for to each house record in the training data set.
2. A non-linear *Boosting Trees* machine learning model is used to predict housig prices based on non-location variables (house size and bedrooms) together with the location variables developed in the first step. The median prediction error based on withheld dta set that was not used to train the model is @@@@. A linear regression model with the same variables has an @@@% higher error. To ensure that the predictions are compatible with 2026 while the underlying data were from 2026 all predicted prices were multiplied with a factor of @@@ which is the increase of the Case-Shiller house price index from @@@ too@@@ (source:())
3. SHAP-Values, a game-theory based algorithm is used to quantify the impact for each of the variables related to a benchmark (the average house value in Los Angeles County). The sum of theses impact factors explains the difference between the average house price in LA and the predicted price of an individual home.

2 Data

The following analysis is based on the *Secured Basic File Abstract* dataset from the [Los Angeles County Assessor Office](#). This core dataset was then enriched with location based data based on the property addresses.

2.1 Property and Sales Records

The original core dataset for the analysis included 2,423,829 sales from 6/1925 to 4/2024 with variables such as sales dat, sales amount, square footage, year build, and number of bedrooms.

The dataset was filtered to include only house sales after the beginning of the year 2023 with a minimum price of \$200,000 and a maximum price below \$3,000,000. The lower limit was used to make sure that the underlying transaction was done at an *Arms length*.¹ The upper limit was used because estimating high priced houses involves factors that are usually not publicly available.

For similar reasons houses with more than 5,000 square feet (or more than 6 bedrooms) where excluded together with houses with less than 500 square feet. Houses older than 1950 where also excluded because the value of *historic* houses varies significantly based on the condition of the house for which no data were available.

The following variables were considered for the analysis: - Property address - Transaction price in US dollars - Square footage - Number of bedrooms - Year Build

¹An *Arms length* business deal assumes that buyer and seller are unrelated and act in their own best interest. This type of sale has the best chances to reflect a true market price.

The variable *Year Build* was ultimately not used because the correlation to the house price was too low (-0.06; Pearson Correlation Coefficient).

The remaining 31,321 real estate transaction records were enriched with location variables based on the property address.

2.2 Location Based Variables

Property and Sales Records:

Since this dataset contained property address the geo-coding tool from the ESRI's ArcGIS Pro software was used to transform each sales record's property address into latitude/longitude (lat/long) coordinates. These **geo-coded sales data** were then enriched with location data as follows.

Drive Time to the Closest Beach:

The beaches of Los Angeles County and some close by beaches of Orange and Ventura County were geocoded with lat/long coordinates. The Network Analyst extension (Closest Facility) together with a street network layer was then used to find the closest beach in term of drive time during normal traffic from. The drive time to this closest beach was then assigned to each of the *Property and Sales Record*.

School Quality:

School addresses for public high schools were downloaded from [California Department of Education](#) and based on the *County-District-School Code (CDC)* linked to the 2022/23 [California Assessment of Student Performance and Progress \(CAASPP\)](#) file that included the *Mean Performance Score* for each school. Since the school address file already included lat/long coordinates the *Near Tool* from ArcGIS Pro could be used to assign the nearest highschool together with its *Mean Performance Score* to each sales record.

3 Price Estimate

4 Impact Factors

5 Application